

Which Carbon Takes the Proton?

A parameter-free computation of the branch that costs GAD65 its cofactor

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Abstract

Glutamate decarboxylase GAD65 — the major type-1-diabetes autoantigen — destroys its own cofactor as a side reaction of its normal catalytic cycle. After CO₂ departs, the resulting quinonoid intermediate can be reprotonated at either of two carbons two bond lengths apart: at C α , which releases GABA and hands back an intact pyridoxal-5'-phosphate, or at C4', which gives pyridoxamine phosphate, falls out of the active site, and leaves the enzyme in the cofactor-free form implicated in its autoantigenicity. Which carbon is favoured therefore sets the rate of self-inactivation.

We compute that preference with RealQM — a real-space method with no basis set, no exchange–correlation functional and nothing fitted — on an active site taken from the GAD65 crystal structure, with the stereoelectronic alignment read off the measured ring normal rather than imposed. A bare proton is placed at each site in turn. The two configurations carry the same twenty-seven kernels, the same sixty-three electrons and the same grid, and differ in the position of one atom, so that the large common errors cancel — the only construction in which this framework’s energies are trustworthy, and we report nothing outside it.

The preference is for C4' — the destructive branch — and it is electronic in origin: the proton there pays 1.15 Ha more in kernel repulsion and recovers 2.29 Ha more in electronic attraction. The direction is stable under refinement; the magnitude is not, and no magnitude is claimed. The result is consistent with solution chemistry that the calculation was not tuned to reproduce: free PLP with an amino acid transaminates rather than decarboxylates. On this evidence the bare quinonoid prefers to destroy its cofactor, and catalysis is what an enzyme must impose against that preference.

Repeated in GAD67 — the paralogue that keeps its cofactor, whose active site is chemically identical — the measurement gives the same preference and no resolved difference: the shift is the size of the convergence drift and runs the wrong way for the hypothesis it would support. The resolved active sites do not differ in this respect, which leaves the disordered catalytic loop, unplaceable in GAD65 and fully built in GAD67, to carry the difference.

1 The question

Glutamate decarboxylase converts glutamate to GABA using pyridoxal-5'-phosphate. In the accepted mechanism the substrate forms an external aldimine with the cofactor, the C α –COO[−] bond breaks in the orientation perpendicular to the ring that Dunathan’s rule selects, and the electron pair left behind delocalises into the protonated pyridine ring: the quinonoid intermediate, an object bright enough at 500 nm to be followed in stopped-flow.

What happens next is a fork, and it is the subject of this article. A proton arrives, and the conjugated chain offers it two carbons:

protonated at	gives	outcome
$C\alpha$ — the substrate’s carbon	product aldimine	GABA released, PLP regenerated
$C4'$ — the cofactor’s carbon	ketimine $\rightarrow$ PMP	cofactor lost, enzyme left <i>apo</i>

They are 2.37 Å apart on the same π system. One returns the enzyme to service; the other converts its cofactor into something that dissociates.

This matters beyond enzymology because GAD65 is the major type-1-diabetes autoantigen, and its propensity to lose PLP and populate a flexible apo form has been linked to that autoantigenicity. The chain runs: branch preference $\rightarrow$ rate of cofactor loss $\rightarrow$ size of the apo pool $\rightarrow$ (on the standing hypothesis) autoantigenicity. We address the first link only. The later ones belong to enzymology and immunology, and the hypothesis joining them is not ours to test.

2 What can be computed, and what cannot

RealQM represents an N -electron system by N real-valued wave functions on non-overlapping supports in three-dimensional space, with the physics in one functional — Coulomb potential energy without self-repulsion, plus kinetic energy — and nothing fitted. It has no basis set and no exchange–correlation functional, so there is no choice to make that could steer the answer, and no population-analysis convention is needed to say where charge sits: the supports *are* the partition.

Against that stands a limitation established by calibration and worth stating before any result. This framework’s *absolute* energetics are unreliable. Its proton affinity for ammonia comes out at -58 kcal/mol against $+204$ measured — the wrong sign — and stays wrong across nine variants of how the nitrogen is represented. The reason is arithmetic rather than mysterious: a proton affinity differences two systems of *different composition*, one of which has an extra kernel, so the errors do not cancel. The same objection retires the obvious design for the present problem, in which one compares a cofactor-bearing system with a cofactor-free one: those are systems of twenty-four and thirteen atoms, and no amount of refinement makes their difference meaningful.

What does survive is a difference *within one composition*. Two configurations of the same atoms, same electron count, same kernels and same grid share their errors, and the errors subtract. The quantity this article reports is of that kind and only that kind.

3 The calculation

Geometry from the structure. The active site is taken from the GAD65 crystal structure [1], which contains the cofactor and the catalytic lysine as a single covalent residue, LLP396, together with the aspartate that hydrogen-bonds the ring nitrogen at 2.46 Å and a bound molecule of the product. The substrate is grafted at the positions the lysine vacates on forming the external aldimine — imine nitrogen at the crystallographic NZ, $C\alpha$ at CE — and the breaking $C\alpha$ –COO bond is laid along the *measured* ring normal, so that the Dunathan alignment comes out of the structure rather than being imposed: it checks to 0.0° . Bond lengths in the grafted fragment agree with standard values ($C4'$ –N7 1.34, N7– $C\alpha$ 1.49 Å). Hydrogens are added, X-ray seeing none. The nuclei are held throughout; only the charge relaxes.

The probe. A bare nucleus — no electrons of its own, a unit kernel — is a proton, and is how proton transfer has been represented in this framework before. It is placed 2.00 Å from C α on the face that CO₂ vacates, and in the second configuration 2.00 Å from C4' on the face opposite its own hydrogen. The geometry is symmetric between the two: in each the probe is 2.00 Å from its host and 3.88 Å from the other carbon, so neither site is privileged by its placement.

What is held identical. Both configurations carry twenty-seven kernels, sixty-three electrons, total kernel charge sixty-four, one box and one mesh. Position by position, exactly one atom differs between them. The reported quantity is

$$\Delta = E(\text{proton at } C\alpha) - E(\text{proton at } C4'),$$

and positive Δ means C4' is preferred.

4 Result

steps	$E(C\alpha)$ (Ha)	$E(C4')$ (Ha)	Δ (Ha)	residuals
2000	-6.968	-8.208	+1.241	2.08 / 2.04
4000	-7.364	-8.499	+1.136	0.52 / 0.53

Δ is positive at both step counts: **the proton prefers C4'**, the branch that destroys the cofactor.

The preference is electronic, and it wins against the nuclei. The kernel–kernel energy the proton pays is computable in closed form from the geometry, and it runs the other way: 6.98 Ha at C α against 8.13 Ha at C4', because C4' sits nearer the ring and so nearer more kernels. Subtracting,

contribution	Ha
kernel–kernel, favouring C α	-1.15
electronic, favouring C4'	+2.29
net, favouring C4'	+1.14

So the result is not an electrostatic accident of where the nuclei sit. The proton is drawn to C4' by the electrons, against a nuclear term that opposes it by half as much.

How far the cancellation goes, measured rather than assumed. Between the two step counts the individual energies moved by 0.40 and 0.29 Ha while Δ moved by 0.105 — a cancellation of 73.5 per cent. That is good enough to fix the sign, the drift being 66 kcal/mol against a difference of 713, and not good enough to fix the value. It is also markedly worse than the 99.5 per cent obtainable when two configurations are related by an exact symmetry, which C α and C4' are not. **We therefore report a direction and a mechanism, and no number.**

A consistency check the calculation was not tuned to pass. PLP and an amino acid in solution, with no enzyme present, undergo transamination readily — which is protonation at C4'. If the intrinsic preference were for C α , free PLP would decarboxylate instead. The computed preference agrees with the solution chemistry, and nothing in the model was adjusted to make it do so.

5 Scope

- **A direction, not a rate.** A branch *ratio* requires barriers and a rate theory. A minimisation over static densities contains no time and can supply neither. Nothing here predicts how often GAD65 inactivates itself.
- **One elementary step.** The catalytic cycle has six. Transaldimination, reprotonation and product release are not modelled.
- **Almost no enzyme.** The crystal supplies the geometry, and Asp364 is present, but there is no protein beyond it, no solvent, and — importantly — no Lys396, which is the actual proton donor. The enzyme’s control over this branch is exerted by *positioning* that donor, and a free proton measures the intrinsic preference the enzyme works against, not the enzymatic outcome.
- **Not a first.** Quantum-mechanical/molecular-mechanical work has treated the equivalent C α -to-C4’ transfer in a related PLP enzyme, with explicit lysine shuttle and free-energy profiles, which is more than is offered here. What is new is the absence of fitted content: no functional was chosen and no population convention was needed.

6 The isoform comparison

GAD67, the paralogue, sheds its cofactor far more rarely, and its structure was solved alongside GAD65’s [1]. The two active sites are chemically identical — the same aspartate, histidine, leucine, threonine, asparagine and alanine, the numbering shifted by nine — so setting them side by side is a difference of *geometry at fixed composition*, which is the construction used above. The GAD67 site was built by the same route from its own structure, placed on the same box and mesh, and the same two-position measurement made in it.

	2000 steps	4000 steps	drift
GAD65, Δ (kcal/mol)	+779	+713	66
GAD67, Δ (kcal/mol)	+816	+768	48
isoform shift	+37	+55	

Both prefer C4’. The shift between them, 55 kcal/mol, is the same size as the 48–66 by which each difference is still moving between step counts, and it grows with refinement rather than settling. **It is not resolved, and we do not report an isoform difference.**

Two things make that reading firmer than a bare failure to resolve. The shift runs the *wrong way* for the hypothesis it would support: GAD67, which loses its cofactor less, comes out preferring the destructive branch slightly more. And it was foreseeable from the structures — the two cofactor geometries agree within 0.01 Å at every bond, the sole difference being the aspartate–N1 hydrogen bond at 2.76 Å against 2.46, and that distance is itself uncertain, 1.2 standard deviations against a combined coordinate error of 0.26 Å with the GAD65 atoms carrying displacement parameters two to three times the GAD67 ones. Driving that distance deliberately across 2.2–3.4 Å moves the branch by under one per cent.

So the resolved active sites do not differ in this respect, and the isoform difference must lie where the structures plainly do differ. GAD65’s catalytic loop is disordered — nine of its residues cannot be placed at all — where GAD67’s is fully built; GAD65’s mean atomic displacement parameter is

52.6 against GAD67's 25.6, at the same 2.30 Å resolution. A calculation with its nuclei clamped cannot represent disorder and makes no attempt to. What it can do, and here does, is exclude the alternative: the difference between these isoforms is not in the chemistry of their active sites as crystallised, which leaves the mobile loop to carry it. That is the standing structural explanation, reached here by elimination rather than by analogy.

References

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